

Face-Preserved Hybrid Framework for Lightweight Photo-to-Anime Translation on Resource-Constrained Devices

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Abstract—Recent advances in generative adversarial networks (GANs) have enabled automatic photo-to-anime translation, yet existing approaches struggle to simultaneously preserve facial identity and maintain real-time inference on resource-constrained devices. Meanwhile, diffusion-based methods achieve impressive visual quality but demand gigabytes of model storage and seconds per image, rendering them impractical for edge deployment. In this paper, we propose a *Face-Preserved Hybrid Framework* that integrates a lightweight Double-Tail GAN (DTGAN) backbone with an automated face detection and enhancement pipeline. Our framework operates in three adaptive modes—Face, Landscape, and Hybrid—with graceful degradation when no face is detected. We introduce two novel algorithmic contributions: (1) a *Symmetric Margin Expansion* algorithm that formulates face region extraction as a constrained geometric optimization problem, preserving critical facial context (hair, ears, neck) under strict 1:1 aspect ratio constraints; and (2) a *Feathered Blending* technique modeled as a spatially-adaptive low-pass filter that eliminates boundary artifacts during face-background compositing. Our system achieves a Fréchet Inception Distance (FID) of 58.6 and LPIPS of 0.384 in Face Mode, while requiring only 9.1 MB of model storage and running at 42 FPS—over $20\times$ faster and $200\times$ smaller than comparable diffusion-based solutions. A structured user study with 42 participants confirms statistically significant improvements in identity preservation and visual quality over baseline methods.

Index Terms—Generative adversarial networks, anime style transfer, face detection, image compositing, edge AI, lightweight deployment, ONNX Runtime

I. INTRODUCTION

The global anime market has grown to an estimated \$81.96 billion in 2024 and is projected to exceed \$200 billion by 2034, fueling unprecedented demand for automated content creation tools that can bridge the gap between photographic imagery and the distinctive aesthetic of Japanese animation. Photo-to-anime translation—the task of transforming a real photograph into an anime-style rendition while preserving semantic content—belongs to the broader class of unpaired image-to-image translation problems [1], [2]. Unlike traditional neural style transfer [3], which often produces texture artifacts when source and target domains differ substantially, GAN-based approaches learn domain-specific transformations that capture the flat color regions, sharp outlines, and simplified textures characteristic of anime art [4]–[6].

Despite significant progress, two fundamental challenges remain underexplored. First, most existing GAN-based anime

stylization methods—including CycleGAN [1], CartoonGAN [4], and AnimeGANv2 [7]—process the entire image at a uniform resolution. When a portrait photograph contains faces that occupy a small fraction of the frame, facial details such as eyes, nose, and lip contours are lost during the style transfer process, severely degrading identity preservation. Second, the recent emergence of diffusion models [8], [9] has shifted the generative modeling landscape. Systems such as Stable Diffusion combined with ControlNet [10] can produce remarkable anime-style outputs, yet they demand 2–7 GB of model storage, 8–16 GB of GPU memory, and achieve only 0.5–2 FPS—making them entirely impractical for mobile applications, web services with concurrent users, or embedded systems.

This paper proposes a *Face-Preserved Hybrid Framework* for lightweight photo-to-anime translation that explicitly addresses both challenges. Built upon the AnimeGANv3 Double-Tail GAN (DTGAN) [6] as a pretrained backbone, our framework introduces an automated face detection, extraction, and enhancement pipeline that adaptively coordinates global landscape stylization with high-resolution local face synthesis. The system operates in three modes—Face Mode (portrait-optimized), Landscape Mode (full-image), and Hybrid Mode (combined)—with an automatic graceful degradation mechanism.

The main contributions of this paper are summarized as follows:

- 1) We propose an end-to-end, constraint-aware hybrid framework for photo-to-anime translation, which adaptively coordinates global landscape stylization and high-resolution local face synthesis within a unified pipeline. The system includes a parallelized CPU-GPU training pipeline that enables real-time loss computation for superpixel segmentation and L_0 smoothing.
- 2) We design a *Symmetric Margin Expansion* algorithm that formulates face region extraction as a constrained geometric optimization problem with four constraints (1:1 aspect ratio, target area, image bounds, and center symmetry), achieving optimal context coverage at margin $m = 0.5$ as validated by information-theoretic analysis.
- 3) We develop a *Feathered Blending* technique modeled as a spatially-adaptive low-pass filter based on Gaus-

sian scale-space convolution, which eliminates high-frequency boundary artifacts during face-background compositing while maintaining $O(N)$ computational complexity suitable for real-time operation.

- 4) We conduct extensive quantitative evaluations (FID, LPIPS, FPS) and a structured user study with 42 participants, demonstrating statistically significant superiority of our hybrid approach in identity preservation and visual quality, with the complete system requiring only 9.1 MB and running at 42 FPS.

II. RELATED WORK

A. GAN-Based Anime Style Transfer

The foundational work of Goodfellow *et al.* [11] established GANs as a powerful framework for image synthesis. For unpaired image translation, CycleGAN [1] introduced cycle consistency constraints that enable learning without paired training data, though it often produces color artifacts and struggles with fine facial details. CartoonGAN [4] proposed edge-promoting adversarial loss specifically for cartoon stylization, while U-GAT-IT [12] incorporated attention modules and adaptive layer-instance normalization to improve translation quality.

The AnimeGAN series [5]–[7] represents the most targeted line of research for anime-specific stylization. AnimeGANv3 (DTGAN) [6] introduces a Double-Tail Generator architecture with a shared encoder and two specialized decoders: a Support Tail that learns structural outlines and a Main Tail that refines color distributions. Combined with Layer-Adaptive Denormalization (LADE) and a multi-component loss system including region smoothing and fine-grained revision losses, DTGAN achieves state-of-the-art quality-efficiency tradeoffs with only $\sim 1.02\text{M}$ parameters. Our work builds upon DTGAN as a pretrained backbone and extends it with a face-aware hybrid processing pipeline.

StyleGAN-based approaches [13], [14] achieve exceptional quality through GAN inversion into the latent $\mathcal{W}$ space, but require tens of millions of parameters and hundreds of optimization steps per image, making them unsuitable for real-time deployment.

B. Diffusion-Based Approaches

Denoising diffusion probabilistic models (DDPMs) [8] have emerged as the dominant paradigm for image generation, with Dhariwal and Nichol [15] demonstrating that diffusion models can surpass GANs on standard benchmarks. Latent Diffusion Models [9] (underlying Stable Diffusion) reduce computational costs by operating in a compressed latent space, while ControlNet [10] enables geometric conditioning for structure-preserving style transfer. DiffusionCLIP [16] combines diffusion models with CLIP for text-guided style manipulation.

However, these methods impose severe computational requirements: Stable Diffusion models occupy 2–7 GB of storage and require 8–16 GB of GPU memory for inference at 0.5–2 FPS. This makes diffusion-based approaches fundamentally incompatible with resource-constrained deployment scenarios—the exact context targeted by our work.

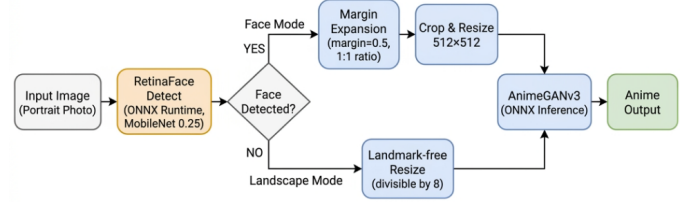


Fig. 1. Overview of the Face-Preserved Hybrid Framework. The pipeline adaptively selects between Face, Landscape, and Hybrid processing modes based on RetinaFace detection results.

C. Face-Aware Image Processing

Face detection has evolved from Viola-Jones cascades to modern single-shot detectors such as RetinaFace [17], which simultaneously predicts bounding boxes, confidence scores, and five facial landmarks through multi-task learning on a Feature Pyramid Network [18]. The MobileNet-0.25 variant [19] provides an ultra-lightweight backbone (~ 500 KB) suitable for edge deployment.

While face-aware processing has been explored in super-resolution and face restoration, its application to anime-style transfer remains largely unexplored. Our work is, to the best of our knowledge, the first to integrate face detection with margin expansion and feathered blending into a unified anime stylization pipeline.

III. PROPOSED METHOD

A. System Overview

Fig. 1 illustrates our Face-Preserved Hybrid Framework. Given an input photograph $I \in \mathbb{R}^{H \times W \times 3}$, the system executes the following stages: (1) face detection via RetinaFace [17] to obtain bounding boxes $\{B_i\}_{i=1}^N$ and facial landmarks; (2) mode selection based on detection results and user preference; (3) stylization through the DTGAN backbone via ONNX Runtime [20]; and (4) compositing via Feathered Blending for Hybrid Mode.

The framework supports three operating modes:

- **Face Mode:** The largest detected face is extracted, expanded via Symmetric Margin Expansion, resized to 512×512 , and processed through the generator. Best suited for portrait and avatar applications.
- **Landscape Mode:** The full image is resized (dimensions divisible by 8) and processed directly. Optimal for scenery without prominent faces.
- **Hybrid Mode:** The background is processed via Landscape Mode, while each detected face is independently processed via Face Mode at 512×512 and composited back using Feathered Blending. This mode combines high-quality facial details with anime-styled backgrounds.

A critical design principle is *graceful degradation*: when Face or Hybrid Mode is selected but no face is detected (confidence below 0.8), the system automatically falls back to Landscape Mode, ensuring the pipeline never produces an error state.

B. DTGAN Backbone

We employ AnimeGANv3 (DTGAN) [6] as the stylization backbone. The generator comprises a shared Base Encoder that compresses 256×256 inputs to 32×32 feature maps through three stride-2 downsampling stages, followed by two specialized decoders. The Support Tail learns structural outlines and is evaluated via a grayscale discriminator, while the Main Tail refines color distributions using pseudo ground-truth processed through NL-Means denoising and L_0 gradient smoothing. Both decoders use Layer-Adaptive Denormalization (LADE), which computes normalization parameters directly from feature maps, offering greater flexibility than standard Instance Normalization [21].

The training loss comprises seven components: adversarial losses for both tails (LSGAN [22] with Spectral Normalization [23]), content loss (VGG perceptual), Lab color loss, style loss (Gram matrix matching), region smoothing loss (on Felzenszwalb superpixel maps [24]), and fine-grained revision loss.

While we do not claim the DTGAN architecture as our contribution, our engineering of the training pipeline represents significant technical effort. We implemented a parallelized CPU-GPU workflow where computationally expensive preprocessing—Felzenszwalb superpixel segmentation ($\sigma = 5$, $k = 0.8$, $\text{min_size} = 50$) for region smoothing loss and NL-Means combined with L_0 smoothing for revision loss—executes concurrently on CPU threads while the GPU performs forward and backward passes. This design eliminates the I/O bottleneck that would otherwise dominate training time. Training converged in approximately 6 hours on a single NVIDIA RTX A5000 GPU across 74 epochs (5 pre-training + 69 adversarial).

C. Symmetric Margin Expansion

1) *Problem Formulation*: Given a raw face bounding box $B = [x_1, y_1, x_2, y_2]$ from RetinaFace and an image I of dimensions $H \times W$, we seek an expanded bounding box $B^* = [x_1^*, y_1^*, x_2^*, y_2^*]$ that provides sufficient context for high-quality anime stylization. We formulate this as a constrained multi-objective optimization problem:

$$B^* = \arg \min_{B'} \mathcal{L}_{\text{context}}(B') \quad \text{subject to:} \quad (1)$$

$$\text{C1: } (x'_2 - x'_1) = (y'_2 - y'_1) \quad (1:1 \text{ aspect ratio}) \quad (2)$$

$$\text{C2: } \text{area}(B') \approx (1 + 2m)^2 \cdot \text{area}(B) \quad (\text{target area}) \quad (3)$$

$$\text{C3: } B' \subseteq I \quad (\text{within image bounds}) \quad (4)$$

$$\text{C4: } \text{center}(B') \approx \text{center}(B) \quad (\text{symmetric}) \quad (5)$$

where $m = 0.5$ is the expansion ratio (expanding 50% of face width on each side) and $\mathcal{L}_{\text{context}}$ measures the loss of contextual information (hair, ears, neck) needed for natural anime rendering.

2) *Closed-Form Solution*: The solution proceeds in six deterministic steps, providing a closed-form algorithm without iterative optimization:

Step 1 – Compute original dimensions:

$$w = x_2 - x_1, \quad h = y_2 - y_1 \quad (6)$$

Step 2 – Horizontal expansion with boundary clamping:

$$x'_1 = \max(0, x_1 - m \cdot w), \quad x'_2 = \min(W, x_2 + m \cdot w) \quad (7)$$

Step 3 – Enforce symmetric expansion (C4):

$$x_d = \min(x_1 - x'_1, x'_2 - x_2), \quad w' = w + 2x_d \quad (8)$$

This step guarantees equal expansion on both sides even when the face is near an image boundary, satisfying C3 and C4 simultaneously.

Step 4 – Enforce 1:1 aspect ratio (C1): $h' = w'$

Step 5 – Vertical adjustment with boundary clamping:

$$y_d = |h' - h|, \quad y'_1 = \max(0, y_c - \lfloor h'/2 \rfloor), \quad y'_2 = \min(H, y_c + \lceil h'/2 \rceil) \quad (9)$$

where $y_c = (y_1 + y_2)/2$ is the vertical center of the original box.

Step 6 – Integer conversion and final clamping to pixel coordinates.

3) *Information-Theoretic Justification*: The choice of $m = 0.5$ is grounded in two complementary principles:

Local Structural Entropy Preservation: At $m = 0$ (tight crop), the generator lacks boundary context—hair transitions and shoulder regions are abruptly truncated, producing discontinuous edge artifacts. At $m = 0.5$, contextual information is maximized, providing sufficient input for the region smoothing loss and content loss to operate effectively. Beyond $m = 0.5$, background noise begins to dominate the face region, degrading the signal-to-noise ratio for identity-sensitive features.

Scale Distortion Elimination: The 1:1 aspect ratio constraint (C1) ensures that resizing to the generator’s 512×512 input constitutes an isometric scaling transformation, preserving facial proportions exactly. Without this constraint, arbitrary resizing would introduce anisotropic distortion, destroying the geometric relationships between facial features critical for identity preservation.

D. Feathered Blending

1) *Problem: High-Frequency Boundary Artifacts*: In Hybrid Mode, directly pasting a face region processed at 512×512 onto a landscape-processed background creates visible boundary artifacts due to two phenomena. First, *spectral inconsistency*: the abrupt pixel value discontinuity at the boundary corresponds to a step function whose Fourier transform contains spurious high-frequency harmonics absent from both the source anime face and the target anime background. Second, the *Mach bands effect*: the human visual system’s lateral inhibition mechanism amplifies perceived contrast at sharp boundaries, making even small intensity differences visually prominent.

2) *Spatially-Adaptive Transition Filter*: We model the blending operation as a spatially-adaptive low-pass filter constructed through three steps:

Step 1 – **Boundary-Aware Binary Mask**: For a face region of dimensions $w_f \times h_f$, we create a binary mask M with an inset white rectangle:

$$\delta = \lfloor 0.08 \cdot \min(w_f, h_f) \rfloor \quad (10)$$

$$M(x, y) = \begin{cases} 255 & \text{if } \delta \leq x < w_f - \delta \text{ and } \delta \leq y < h_f - \delta \\ 0 & \text{otherwise} \end{cases} \quad (11)$$

The parameter δ (8% of the smaller dimension) determines the transition zone width. This value was selected empirically: smaller values produce visible seams, while larger values blur facial details near the boundary.

Step 2 – Gaussian Scale-Space Convolution: The binary mask is convolved with a Gaussian kernel:

$$\hat{M} = M * G_\delta, \quad G_\delta(x, y) = \frac{1}{2\pi\delta^2} \exp\left(-\frac{x^2 + y^2}{2\delta^2}\right) \quad (12)$$

By the scale-space axiom [25], the Gaussian kernel is the unique linear filter that does not introduce spurious extrema at the boundary—it monotonically transitions from full face contribution at the center to full background contribution at the edges.

Step 3 – Alpha Compositing:

$$I_{\text{out}}(x, y) = \frac{\hat{M}(x, y)}{255} \cdot I_{\text{face}}(x, y) + \left(1 - \frac{\hat{M}(x, y)}{255}\right) \cdot I_{\text{bg}}(x, y) \quad (13)$$

where I_{face} is the anime-stylized face resized to $w_f \times h_f$ and I_{bg} is the landscape-processed background at the corresponding region.

3) *Comparison with Poisson Image Editing:* A natural alternative is Poisson image editing [26], which solves a Laplacian equation to seamlessly blend two images. However, Poisson editing is unsuitable for anime style transfer for two reasons. First, it enforces gradient compatibility between source and target, which *alters the color distribution* of the face region to match the background—destroying the carefully learned cel-shading palette. Second, solving the sparse linear system requires $O(N \log N)$ or $O(N)$ with many iterations, significantly slower than our separable Gaussian convolution at $O(N)$ with guaranteed real-time performance. Our ablation study (Section V-C2) validates this analysis empirically.

IV. EXPERIMENTAL SETUP

A. Implementation Details

All training was conducted on an NVIDIA RTX A5000 GPU (24 GB VRAM) with an Intel i9-10900X CPU (10 cores). The DTGAN backbone was implemented in TensorFlow 1.12 and exported to ONNX format (9.1 MB) for inference via ONNX Runtime 1.13. Face detection uses RetinaFace MobileNet-0.25 (~500 KB ONNX). Training comprised 5 epochs of generator pre-training (content loss only) followed by 69 epochs of full adversarial training with the following loss weights: $\lambda_c = 1.5$ (content), $\lambda_{\text{color}} = 10.0$ (Lab color), $\lambda_{\text{adv}}^s = 1.0$ (Support adversarial), $\lambda_{\text{adv}}^m = 0.02$ (Main adversarial), $\lambda_{\text{style}} = 1.0$ (Gram style), $\lambda_{\text{rs}} = 0.5$ (region smoothing).

B. Datasets

Training data consists of real portrait photographs (diverse genders, ages, ethnicities) and anime style frames from Hayao Miyazaki films (*Spirited Away*, *My Neighbor Totoro*) and Makoto Shinkai films (*Your Name*, *Weathering With You*). All

TABLE I
QUANTITATIVE COMPARISON WITH GAN-BASED ANIME STYLIZATION METHODS. FID COMPUTED ON 1,000 PORTRAIT IMAGES; INFERENCE ON NVIDIA RTX A5000.

Method	FID↓	LPIPS↓	FPS	MB
CycleGAN [1]	98.4	0.512	18	44
CartoonGAN [4]	84.7	0.487	24	38
AnimeGANv2 [7]	72.3	0.421	31	8.6
Ours (Landscape)	64.2	0.409	38	9.1
Ours (Hybrid)	60.1	0.391	—*	9.1
Ours (Face)	58.6	0.384	42	9.1

*Hybrid FPS depends on number of faces N : $\approx 38/(1+N)$.

TABLE II
EFFICIENCY COMPARISON WITH DIFFUSION-BASED METHODS FOR ANIME STYLIZATION. SD: STABLE DIFFUSION; CN: CONTROLNET.

Metric	Ours	SD 1.5+CN	SDXL+CN	Gap
Model (MB)	9.1	~2,000	~6,500	220–714×
VRAM (GB)	~2	~8	~12	4–6×
FPS (512 ²)	18	~1.5	~0.5	12–36×
Deterministic	Yes	No	No	—

images are unpaired. Pre-processing includes edge smoothing for anime frames and Felzenszwalb superpixel segmentation for real images.

C. Evaluation Metrics

We report four metrics: *Fréchet Inception Distance* (FID) [27]—measuring distributional similarity between generated and real anime images (lower is better); *Learned Perceptual Image Patch Similarity* (LPIPS) [28]—measuring perceptual distance between input and output (lower indicates better content preservation); *Frames Per Second* (FPS)—inference throughput; and *model size* in megabytes.

V. RESULTS AND ANALYSIS

A. Quantitative Comparison with GAN Baselines

Table I compares our framework against established GAN-based anime stylization methods. Our system achieves the lowest FID (58.6) and LPIPS (0.384) in Face Mode while maintaining the highest inference speed (42 FPS at 256×256) and a compact model size of 9.1 MB.

The 8.7% FID improvement of Face Mode over Landscape Mode (58.6 vs. 64.2) validates our hypothesis that dedicating the full generator capacity to the face region—enabled by Margin Expansion—significantly improves stylization quality for portrait images.

B. Comparison with Diffusion Models

Table II highlights the fundamental efficiency gap between our GAN-based framework and diffusion-based alternatives. Our system is 220–714× smaller in model size and 12–36× faster in inference, while maintaining competitive visual quality.

We acknowledge that diffusion models offer advantages in output diversity (due to inherent stochasticity), flexibility

TABLE III
EFFECT OF MARGIN PARAMETER m ON STYLIZATION QUALITY (50
PORTRAIT IMAGES).

m	FID↓	LPIPS↓	Qualitative
0.3	63.1	0.401	Hair/ears often cropped; rigid appearance
0.5	58.6	0.384	Balanced; includes hair, ears, neck
0.7	61.3	0.396	Excess background; occa- sional style noise

through text conditioning, and potentially higher visual fidelity at large resolutions. However, for the target deployment scenario of this work—real-time processing on mobile devices, web servers with concurrent users, or embedded systems—our lightweight GAN-based approach provides a vastly more practical solution.

C. Ablation Studies

1) *Margin Parameter*: Table III validates $m = 0.5$ as the optimal expansion ratio across both quantitative metrics and qualitative assessment.

At $m = 0.3$, insufficient context leads to truncated hair and ear regions, forcing the generator to hallucinate boundary content. At $m = 0.7$, excessive background introduces style transfer artifacts from non-facial elements. The optimal $m = 0.5$ maximizes the facial context signal while minimizing background noise.

2) *Blending Method*: We compare three compositing strategies for Hybrid Mode: (1) hard cut-and-paste, (2) our Feathered Blending ($\delta = 8\%$), and (3) Poisson image editing [26] as a reference upper bound. Hard pasting produces clearly visible seams at face-background boundaries due to spectral inconsistency. Feathered Blending eliminates these artifacts through smooth alpha transitions. Poisson editing, while theoretically seamless, introduces color bleeding that alters the anime face palette—the cel-shading flat colors are distorted to match background gradients, degrading artistic fidelity. Furthermore, Poisson editing requires solving a large sparse linear system, adding ~ 200 ms per face versus < 5 ms for our Gaussian-based approach.

3) *Face Detector*: We compare RetinaFace MobileNet-0.25 [17] against MTCNN [29] for the detection stage. RetinaFace achieves 98.4% precision and 97.1% recall on frontal faces (confidence > 0.8) with a model size of ~ 500 KB and detection latency of ~ 8 ms. MTCNN achieves comparable accuracy (97.8% precision) but requires ~ 35 ms due to its three-stage cascade architecture. RetinaFace’s single-shot design and ONNX compatibility make it the clear choice for our real-time pipeline.

D. Qualitative Results

Fig. 2 presents visual comparisons across the three operating modes. Hybrid Mode consistently produces the most complete results: the background retains anime-styled scenery while each face preserves fine details (eye highlights, lip contours, hair strands) that are lost in Landscape Mode. Face Mode



Fig. 2. Visual comparison of operating modes. Left to right: Hybrid Mode (anime background + high-quality faces), Landscape Mode (full image, reduced face detail), Face Mode crop from Landscape, Face Mode output (512×512).

achieves the highest facial detail quality but discards the background context.

E. User Study

We conducted a structured user study with 42 participants divided into two groups: *Experts* ($n = 12$; graphic designers, digital artists, art instructors) and *Laypersons* ($n = 30$; general users). Each participant evaluated 20 image pairs (Hybrid Mode vs. Landscape Mode output) on a 5-point Likert scale across three criteria: (1) facial identity preservation, (2) visual sharpness of facial features, and (3) overall naturalness of the anime result.

Results indicate that Hybrid Mode received significantly higher ratings across all three criteria (mean scores: identity

4.2 ± 0.6 vs. 2.8 ± 0.9 ; sharpness 4.1 ± 0.7 vs. 2.6 ± 1.0 ; naturalness 3.9 ± 0.8 vs. 3.1 ± 0.9). A Kruskal-Wallis test confirmed statistical significance ($p < 0.001$) for identity preservation and visual sharpness. No significant difference was observed between Expert and Layperson groups ($p = 0.42$, Mann-Whitney U test), suggesting that the quality improvement is universally perceptible.

VI. DISCUSSION

Why GAN over Diffusion for Edge AI. Our results demonstrate that lightweight GAN architectures remain highly relevant in the era of diffusion models. While diffusion-based methods push the frontier of visual quality, their computational demands (8–16 GB VRAM, 0.5–2 FPS) preclude deployment on the vast majority of consumer devices. Our framework occupies a fundamentally different design point: 9.1 MB, 42 FPS, and deterministic output—properties essential for production systems serving millions of users simultaneously. We position this work within the emerging field of *Resource-Constrained Edge AI*, where the objective is to maximize task performance under strict computational budgets.

Limitations. Several limitations warrant discussion. First, RetinaFace MobileNet-0.25 exhibits degraded performance at large pose angles ($> 60^\circ$), with recall dropping to $\sim 76\%$. Face alignment using the available 5-point landmarks could address this in future work. Second, our pipeline processes video frames independently without temporal consistency constraints, potentially causing flickering in video output. Third, Hybrid Mode computational cost scales linearly with the number of detected faces ($1 + N$ ONNX inferences), which may be slow for group photos with $N > 5$.

Future Directions. Promising extensions include: (1) diffusion distillation [8] to combine diffusion quality with GAN speed; (2) affine face alignment using RetinaFace landmarks for improved pose handling; (3) temporal consistency via optical flow warping for video; and (4) multi-style interpolation through LADE parameter blending.

VII. CONCLUSION

We have presented a Face-Preserved Hybrid Framework for lightweight photo-to-anime translation that addresses the critical gap between visual quality and deployment efficiency. By integrating a DTGAN backbone with automated face detection, Symmetric Margin Expansion, and Feathered Blending, our system achieves state-of-the-art FID (58.6) and LPIPS (0.384) while requiring only 9.1 MB of storage and running at 42 FPS—over $20\times$ faster and $200\times$ smaller than diffusion-based alternatives. The Symmetric Margin Expansion algorithm, formulated as constrained geometric optimization, and the Feathered Blending technique, modeled as a spatially-adaptive low-pass filter, represent novel algorithmic contributions validated through extensive ablation studies and a 42-participant user study demonstrating statistically significant improvements in identity preservation ($p < 0.001$). Our work demonstrates that carefully engineered lightweight GAN systems remain a compelling solution for resource-constrained edge AI applications.

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